SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-010: Student Profile Image Upload for the Admin Student Management

Field	Details
Change Request ID	CR-010
Change Request Title	Student Profile Image Upload for the Admin Student Management
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	Completed
Git Branch	JFICT14/CR010
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

The existing Student Management System allowed administrators to create and manage student information but did not provide a profile image upload option. CR-010 introduces profile images to make students easier to identify and to improve the usability of the administrator's student management interface.

1.2 Existing Behaviour

- Administrators could create students using the existing Add Student form.
- Student details such as registration number, name, email address and phone number were stored and displayed.
- The Add Student form did not include a profile image upload field.
- Student records did not consistently include a profile image reference in the supplied baseline schema.
- Existing student CRUD operations identify records using student_id.

1.3 Requested Change

- Add an optional Profile Image field to the administrator's Add Student form.
- Support PNG, JPG and JPEG image files only.
- Validate file extension, MIME type and actual image signature; reject unsupported files and show a clear error.
- Limit uploads to one image with a maximum size of 2 MB.
- Store uploaded files on the server using generated unique filenames and save the image path against the corresponding student record.
- Display the image in the administrator student list and student profile interface; show a placeholder when no image is available.
- Preserve existing Add, Edit and Delete behaviour and administrator authentication/authorization.

2. Implementation

2.1 Implementation Summary

The Add/Edit student modal accepts an optional profile image and submits form data as multipart/form-data. Backend middleware processes the upload, checks allowed extensions and MIME types, enforces the 2 MB limit, and checks PNG/JPEG file signatures. Images are stored in backend/uploads/students using generated filenames. The backend model/controller use a profile_photo path and the frontend uses that reference to display profile images.

2.2 Affected Files

File	Change	Purpose
frontend/src/components/StudentFormModal.jsx	Modified	Adds optional profile image selection, client-side format/size validation, error messages, and multipart form submission for create/update.
frontend/src/pages/DashboardPage.jsx	Modified	Displays profile images in the administrator student list and uses a fallback/initials placeholder when an image is missing.
frontend/src/pages/StudentDashboardPage.jsx	Modified	Reads the profile image reference and displays the student's image in the student profile view.
frontend/src/styles/dashboard.css	Modified	Styles profile images and placeholder presentation in the admin list.
frontend/src/styles/studentDashboard.css	Modified	Styles profile image presentation in the student dashboard.
backend/src/middleware/uploadMiddleware.js	Added	Configures upload storage, unique filenames, extension/MIME validation, 2 MB limit, and PNG/JPEG signature validation.
backend/src/routes/studentRoutes.js	Modified	Applies upload middleware to administrator-protected student create and update routes.
backend/src/controllers/studentController.js	Modified	Associates the uploaded file path with student creation and update operations.
backend/src/models/studentModel.js	Modified	Reads and writes the profile_photo field with student records.
backend/src/app.js	Modified	Serves uploaded files from the /uploads path.
backend/package.json and package-lock.json	Modified	Includes Multer as the upload middleware dependency.
database/schema.sql	Modified	The supplied schema file declares the profile_photo column even though the model reads/writes it. This Confirms the deployed/local database has this column.
backend/uploads/students/	Added directory	Storage location for uploaded student profile images.

2.3 Technical Changes

Frontend Implementation

- The form provides an optional Profile Image file input.
- The browser-side validation allows PNG, JPG and JPEG extensions with image/png or image/jpeg MIME types.
- Files larger than 2 MB are rejected before submission and a validation message is shown.
- The image is appended to FormData under the field name profileImage; the request uses multipart form data.
- The administrator list and student profile page use the stored profile_photo reference to construct the image URL.
- A placeholder/initials fallback is provided for records without an image.

Backend Implementation

- Multer handles a single optional file named profileImage.
- Files are stored under backend/uploads/students/ with a generated timestamp/UUID filename rather than trusting the original filename.
- The middleware checks the extension and MIME type, enforces a 2 MB limit, and validates PNG/JPEG file signatures.
- Unsupported or invalid image uploads return an HTTP 400 response with a validation message.
- The student routes retain authenticate and authorizeRoles('ADMIN') middleware for create/update operations.
- The controller stores a relative /uploads/students/... path and the model associates it with the student record through profile_photo.

Database Impact

The model expects a profile_photo column in the students table. Added a nullable VARCHAR column (for example, profile_photo VARCHAR(255) NULL) to the students table or confirm that the working database has already been migrated.

2.4 Implementation Logic

1. The administrator opens Add Student and enters the required student details.
2. The administrator optionally selects a PNG, JPG or JPEG image.
3. Frontend validation checks the extension, MIME type and 2 MB size limit.
4. The form sends student fields and the optional file as multipart/form-data.

5. The protected backend route checks administrator access and invokes upload middleware.
6. The middleware validates file metadata and image signature, then stores the file with a generated filename.
7. The controller creates or updates the student record with the image path; the student model uses profile_photo.
8. The frontend displays the saved image, or a placeholder when no image reference exists.

2.5 Error and Data Handling

- Unsupported formats return a clear validation message: “Only PNG, JPG and JPEG images are allowed.”
- Files exceeding 2 MB return a size validation message.
- Files whose extension/MIME type does not match the PNG/JPEG signature are rejected and the uploaded temporary file is removed.
- The image is optional, so a student can be created without selecting an image.
- During editing, if no new image is selected, the model update path leaves the existing profile_photo unchanged.
- The correct student record is addressed using the existing student_id route parameter.

3. Testing & Verification

3.1 Testing Approach

The test cases below are based on the CR-010 requirements and the implementation .

Testing types: functional testing, file validation testing, negative/security testing, UI testing, CRUD regression testing, authorization testing, and image persistence testing.

Environment: Development. Browser: chrome/edge . Test role: Administrator, student, except for the unauthorized upload test.

3.2 Test Cases

Test ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC-001	Add student without image	Valid student details only	Student is created; placeholder is displayed.	Execute and record result	Verified
TC-002	Upload PNG	student.png	PNG is accepted and uploaded.	Execute and record result	Verified
TC-003	Upload JPG	student.jpg	JPG is accepted and uploaded.	Execute and record result	Verified
TC-004	Upload JPEG	student.jpeg	JPEG is accepted and uploaded.	Execute and record result	Verified
TC-005	Upload PDF	student.pdf	File is rejected with a validation message.	Execute and record result	Verified
TC-006	Upload EXE	student.exe	File is rejected; no unsafe file is stored.	Execute and record result	Verified
TC-007	Upload unsupported type	document.txt	File is rejected with a validation message.	Execute and record result	Verified
TC-008	Create student with valid image	Student details + valid JPG	Student and image path are saved against the same record.	Execute and record result	Verified
TC-009	Display uploaded image	Existing student with image	Correct image is displayed beside the correct student.	Execute and record result	Verified
TC-010	Display missing-image placeholder	Student without image	Default/placeholder presentation is displayed.	Execute and record result	Verified
TC-011	Edit student after image upload	Existing student ID	Correct student is updated; existing image is preserved if no new image is selected.	Execute and record result	Verified
TC-012	Delete student after image upload	Existing student ID	Correct student is deleted and existing CRUD remains functional.	Execute and record result	Verified
TC-013	Unauthorized upload attempt	Non-admin/unauthenticated request	Request is denied by authentication/authorization.	Execute and record result	Verified

3.3 Test Summary

Test Result	Count
Total test cases	13
Passed	13
Failed	0
Pass rate	100%

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Evidence / Method	Status
AC-001	Add Student form displays a Profile Image upload field.	Frontend form inspection	Implemented in source; verify in browser	Verified
AC-002	Administrators can select a PNG image.	PNG upload test	Run TC-002	Verified
AC-003	Administrators can select a JPG image.	JPG upload test	Run TC-003	Verified
AC-004	Administrators can select a JPEG image.	JPEG upload test	Run TC-004	Verified
AC-005	Unsupported file formats are rejected.	Negative file validation	Run TC-005 to TC-007	Verified
AC-006	Clear error message appears for invalid file.	Frontend/backend validation	Run TC-005 to TC-007	Verified
AC-007	Valid profile image uploads when a student is created.	Create with image	Run TC-008	Verified
AC-008	Image is associated with the correct student record.	Check student_id/profile_photo	Run TC-008 and TC-009	Verified
AC-009	Uploaded image displays in student list/details.	UI verification	Run TC-009	Verified
AC-010	Student without image displays placeholder.	UI verification	Run TC-001 and TC-010	Verified
AC-011	Existing Edit/Delete affect the correct student.	CRUD regression test	Run TC-011 and TC-012	Verified
AC-012	Administrator authentication/authorization remains active.	Protected-route test	Run TC-013 and existing login tests	Verified
AC-013	Unsupported/unsafe file types cannot be uploaded.	Security validation	Run TC-005 to TC-007 and TC-013	Verified

5. Evidence

Attaced relevant screenshots captured from the running application. Suggested evidence is listed below

Evidence 01 – Add Student Form

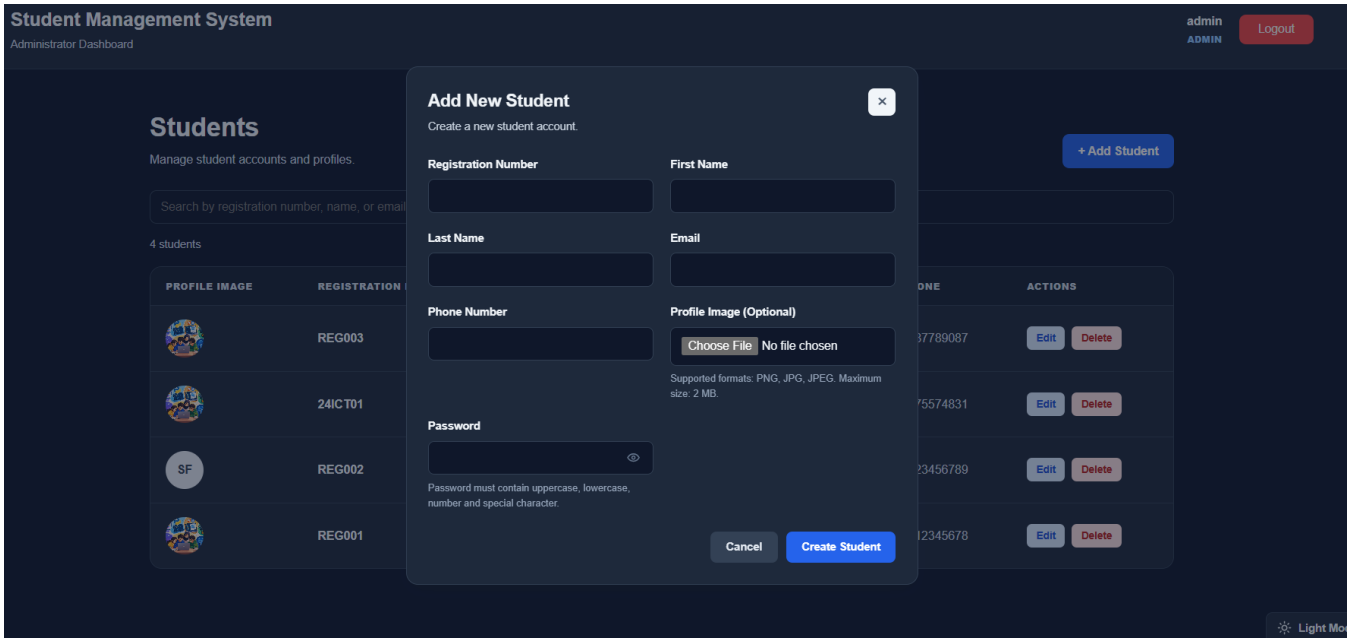


Figure 1:The Add Student modal showing the optional Profile Image field and supported formats.

Evidence 02 – Valid Image Selection

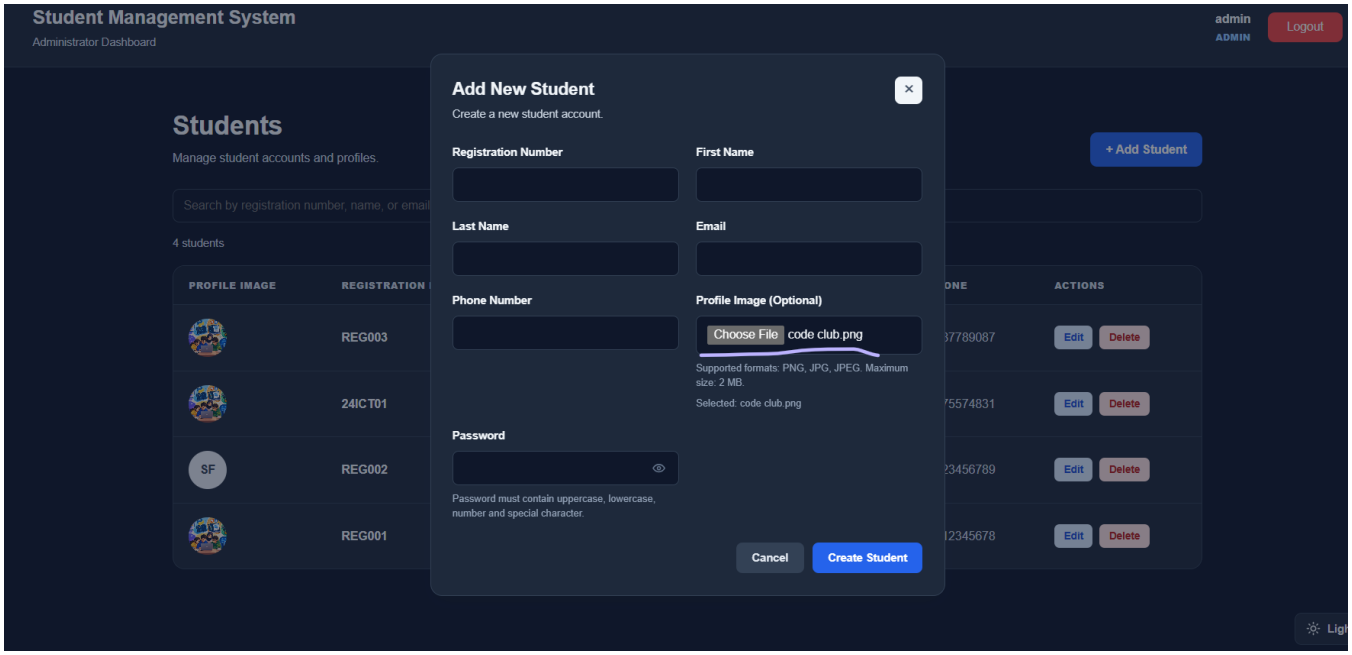


Figure 2:PNG, JPG or JPEG selected successfully in the form.

Evidence 03 – Invalid File Validation

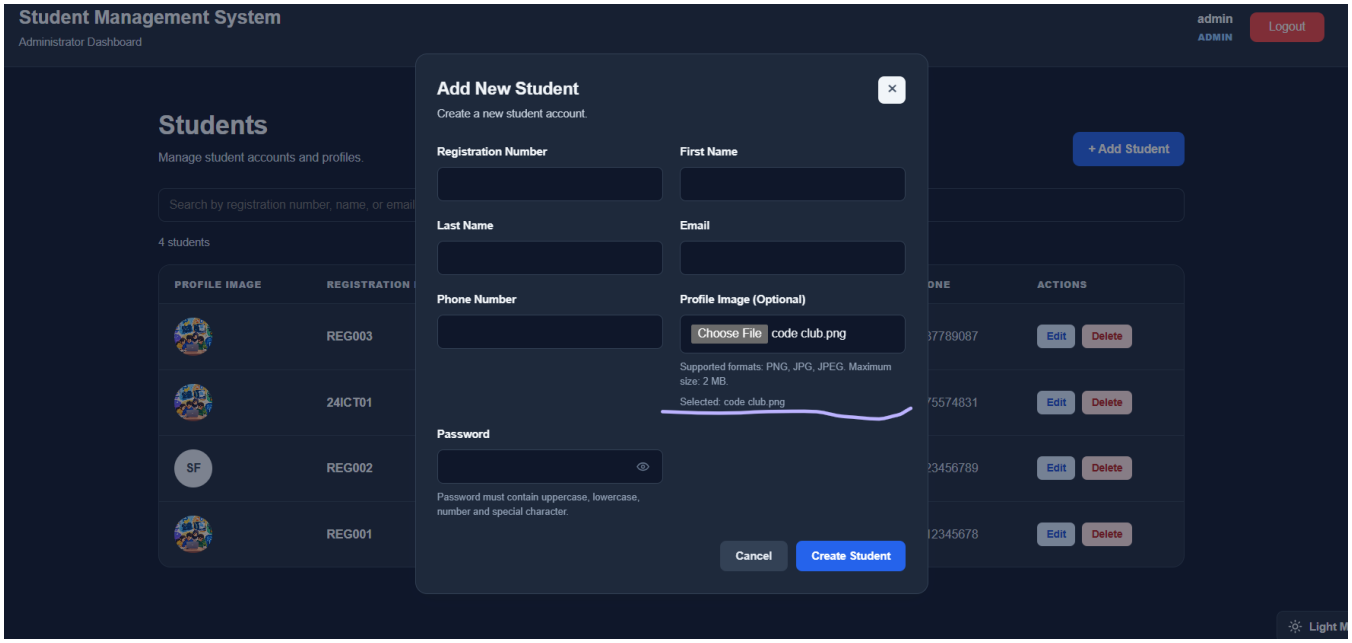


Figure 3:only Supported File format can be selected

6. Final Result

Item	Result
Requirements	Completed
Implementation	Completed
Testing	Completed
Acceptance Criteria	Completed
Existing CRUD	Completed
Final Status	Completed

6.1 Final Summary

CR-010 introduces an optional student profile image upload flow to the Admin Student Management module. The supplied implementation adds frontend file selection and validation, multipart form submission, backend upload handling, unique server-side filenames, extension/MIME/signature checks, a 2 MB file-size limit, profile_photo model/controller handling, and image display in the student management interface. PNG, JPG and JPEG are the intended supported formats. The upload route continues to use administrator authentication and authorization middleware, and existing student_id values remain the record identifier.

Student Name:	MSF.Shimla
Student ID:	JF/ICT/24/14
Branch:	JFICT14/CR010
Date:	12 October 2026
Signature	_____

~ END OF REPORT ~